

Analysis and design of imaging Fabry-Perot interferometers for measurement of Rayleigh-Brillouin scattering spectra in gas flows

LI Tao, LI Gao-ping, WU Lei, YUAN Liang, YU Bing, WANG Xue-xin, YANG Hong-ru,
WU Li-peng, WANG Le, YU Dong-yu, YIN Wan-hong
Optical Metrology Laboratory, Xi'an Institute of Applied Optics, Xi'an 710065, P. R. China

ABSTRACT

Laser Rayleigh-Brillouin scattering is a powerful diagnostic tool for the study of gas flow properties. It provides an effective method for non-intrusive measurement of density, temperature and velocity in the gas flow. The received scattered laser light power is proportional to the gas density, the linewidth of the Rayleigh-Brillouin scattering spectrum is related to the gas temperature, and the Doppler frequency shift of the peak of the Rayleigh-Brillouin scattering spectrum is related to the gas velocity. The Rayleigh-Brillouin scattering spectrum can be measured by a Fabry-Perot interferometer operated in the imaging mode where an intensified CCD camera is frequently used to record the interference patterns of the Fabry-Perot interferometer. The Rayleigh-Brillouin scattering spectrum is then reconstructed from the measured data deconvolved with the Fabry-Perot instrument function. In this paper, the analysis and design of an imaging Fabry-Perot interferometer for the measurement of the Rayleigh-Brillouin scattering spectrum in the gas flow is presented. Some factors that limit the performance of the imaging Fabry-Perot interferometer are analyzed and discussed.

Keywords: Rayleigh-Brillouin scattering, scattering spectrum, Fabry-Perot interferometer, gas flow

1. INTRODUCTION

Laser Rayleigh-Brillouin scattering provides an effective method for non-intrusive measurement of gas flow properties [1-13]. The received scattered laser light power is proportional to the gas density, the linewidth of the Rayleigh-Brillouin scattering spectrum is related to the gas temperature, and the Doppler frequency shift of the peak of the Rayleigh-Brillouin scattering spectrum is related to the gas velocity. The measurement of gas flow's properties relies on the exact knowledge of the spectral profile of the scattered light. The Rayleigh-Brillouin scattering spectra can be acquired by a Fabry-Perot interferometer operated in the imaging mode where an intensified CCD camera is frequently used to record the interference patterns of the Fabry-Perot interferometer [3-13]. The Rayleigh-Brillouin scattering spectrum is then reconstructed from the measured data deconvolved with the Fabry-Perot instrument function. In this paper, the analysis and design of an imaging Fabry-Perot interferometer for the measurement of the Rayleigh-Brillouin scattering spectra in gas flows is presented. Some factors that limit the performance of the imaging Fabry-Perot interferometer are analyzed and discussed.

2. THEORY OF RAYLEIGH-BRILLOUIN SCATTERING

The scattering of light by gases is well known as Rayleigh-Brillouin scattering [14-16]. Rayleigh scattering is the result of the elastic light scattering from gas molecules (or atoms). The internal energy of the gas molecule is unchanged and the frequency (or the spectrum width) of the scattered light is shifted (or broadened) only by the thermal motion of the gas molecules. Brillouin scattering is the result of the inelastic light scattering from the gas pressure fluctuations. The Brillouin scattering can be viewed as a simple process in which the incident light is scattered by acoustic phonons and the frequency of the scattered light is changed associated with this process.

The laser Rayleigh-Brillouin scattering spectrum contains information about the density, temperature, and the bulk velocity of the gas. The intensity of the Rayleigh-Brillouin scattering spectrum is proportional to the gas density. The linewidth of the Rayleigh-Brillouin scattering spectrum is related to the gas temperature. And the Doppler frequency shift between the peak of the laser light spectrum and the Rayleigh-Brillouin scattering spectrum is related to the gas velocity.

The shape of the Rayleigh-Brillouin scattering spectrum is mainly dependent on the pressure and temperature of the gas and the scattering angle. The Rayleigh-Brillouin scattering is usually divided into three different scattering regimes by comparing the molecular mean free path to the wavelength of the incident laser light. A non-dimensional parameter y is usually introduced to describing the scattering regimes [14-16]:

$$y = \frac{1}{2\pi} \frac{\Lambda}{L_m} = \frac{\lambda_L}{4\pi n_g \sin(\theta_s/2)} \frac{p}{\eta_s v_T}, \quad (1)$$

where $\Lambda = \lambda_L / [2n_g \sin(\theta_s/2)]$ is the characteristic scattering wavelength, $L_m = v_T / \alpha_{col}$ is the mean free path of the gas molecules, $\alpha_{col} = p / \eta_s$ is the effective collision frequency, $v_T = \sqrt{2k_B T / m}$ is the thermal speed of the gas molecules, λ_L is the wavelength of the incident laser light, n_g is the refractive index of the gas, θ_s is the scattering angle, η_s is the shear viscosity, k_B is the Boltzmann constant, m is the molecular mass, T is the temperature, and p is the pressure.

For low density gases ($y \ll 1$), the mean free path of the scattering medium (gas flow) is much larger than the wavelength of the incident light, the scattering occurs in the Knudsen regime. In this regime, the profile of the Rayleigh-Brillouin scattering spectrum is approximately a Gaussian function [14-16]:

$$S(k, \omega) = \frac{\sqrt{2\pi}}{k v_T} \exp\{-\omega^2 / (2k^2 v_T^2)\}, \quad (2)$$

where $S(k, \omega)$ is the Rayleigh-Brillouin scattering spectrum, v_T is the thermal speed of the gas molecules, ω is the angle frequency, $k \approx 2k_i \sin(\theta_s/2)$ is the magnitude of the interaction wave vector, and k_i is the magnitude of the incident wave vector.

For dense gases ($y \gg 1$), the mean free path of the scattering gas medium is much smaller than the wavelength of the incident light, the scattering occurs in the hydrodynamic regime. In this regime, the Rayleigh-Brillouin scattering spectrum is composed of three Lorentzian components displaced by the corresponding sound velocity of the scattering medium [14-16]:

$$S(k, \omega) = \frac{\gamma-1}{\gamma} \frac{k^2 / \rho_0 c_p}{\omega^2 + (\kappa k^2 / \rho_0 c_p)^2} + \frac{1}{\gamma} \frac{\Gamma k^2}{(\omega + k v_s)^2 + (\Gamma k^2)^2} + \frac{1}{\gamma} \frac{\Gamma k^2}{(\omega - k v_s)^2 + (\Gamma k^2)^2}, \quad (3)$$

where $\gamma = c_p / c_v$ is the adiabatic index, c_p is the isobaric heat capacity, c_v is the heat capacity at constant volume, ρ_0 is the gas density, κ is the thermal conductivity, Γ is the acoustic wave damping, v_s is the sound velocity, and $\omega = \pm k v_s$ are the Brillouin frequency shifts.

The first term in Eq.(3) is the Rayleigh scattering component, and the other two terms are the Stokes and the anti-Stokes Brillouin scattering components, respectively. The acoustic wave damping is given by [14-16]

$$\Gamma = \frac{1}{2\rho_0} \left[\frac{4}{3} \eta_s + \zeta + \left(\frac{\gamma-1}{\gamma} \right) \frac{k}{c_v} \right], \quad (4)$$

where η_s and ζ are the shear viscosity and the bulk viscosity, respectively.

If the mean free path of the scattering gas medium is comparable to the wavelength of the incident light ($y \sim 1$), the scattering occurs in the kinetic regime. In this scattering regime, the Rayleigh-Brillouin scattering spectrum is a complicated mixture of the Knudsen and the hydrodynamic spectra. The Rayleigh-Brillouin scattering spectrum can be modeled by the TENTI S6 model theoretically [17]. Figure 1 shows the typical Rayleigh-Brillouin scattering spectra of N_2 gas at the different scattering regimes.

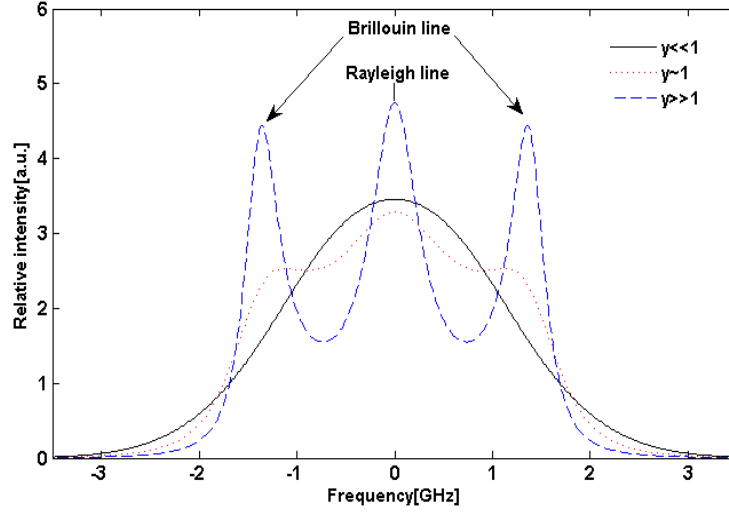


Fig. 1. Rayleigh–Brillouin profiles at the different scattering regimes.

The Rayleigh scattering section can be calculated from

$$\sigma(\nu) = \frac{24\pi^3 \nu^4}{n_{den}^2} \frac{(n_v^2 - 1)^2}{(n_v^2 + 2)^2} F_k(\nu), \quad (5)$$

where $\sigma(\nu)$ is the scattering cross section at the frequency of ν , n_v is the refractive index of the gas medium, $n_{den} = p/(k_B T)$ is the density of gas molecules, and $F_k(\nu)$ is the King correction factor.

The relationship between the area (scattering intensity) of the Rayleigh scattering component and the Brillouin scattering components is known as Landan-Placzek ratio, which is given by [14-16]

$$\frac{2I_B}{I_R} = \frac{1}{\gamma - 1}, \quad (6)$$

where I_R and I_B are the areas of the Rayleigh scattering and the Brillouin scattering components, respectively.

When Rayleigh-Brillouin scattering is used for diagnostics, the scattered light is collected over a limited solid angle which is determined by the collection lens. In this case the receiver light power is derived from the differential scattering cross-section. The differential scattering cross-section can be written as

$$\frac{\partial \sigma}{\partial \Omega} = \frac{3}{8\pi} \sigma(\nu) \sin^2 \phi = \frac{9\pi^2 \nu^4}{n_{den}^2} \frac{(n_v^2 - 1)^2}{(n_v^2 + 2)^2} F_k(\nu) \sin^2 \phi, \quad (7)$$

where $\partial \sigma / \partial \Omega$ is the differential scattering cross-section and ϕ is the scattering polarization angle.

3. IMAGING FABRY-PEROT INTERFEROMETERS FOR MEASUREMENT OF RAYLEIGH-BRILLOUIN SCATTERING SPECTRA IN GAS FLOWS

The Rayleigh-Brillouin scattering spectra of the gas flow can be measured by a Fabry-Perot interferometer operated in the imaging mode where an intensified CCD camera is frequently used to record the interference patterns of the Fabry-Perot interferometer. The Rayleigh-Brillouin scattering spectra are then reconstructed from the measured data deconvolved with the Fabry-Perot instrument function.

The imaging Fabry-Perot interferometer is mainly composed of a receive optics, a Fabry-Perot interferometer (Fabry-Perot etalon), and an intensified CCD camera. The receive optics is used to collect and collimate the laser light scattered

by the gas flow. The Fabry-Perot interferometer is a device that relies on the interference of multiple reflected light beams to resolve the incoming laser light spectrum [18-19]. The Fabry-Perot interferometer is composed of two parallel planar mirrors with their reflective surfaces facing each other. Maximum transmission of the Fabry-Perot etalon occurs when the optical path difference (OPD) between each transmitted beam satisfied the equation:

$$OPD = 2nd \cos \theta + \delta = m\lambda, \quad (8)$$

where $OPD = 2nd \cos \theta + \sigma$ is the optical path length difference, n is the refractive index of the medium in the cavity of the Fabry-Perot interferometer, d is the mirror spacing of the Fabry-Perot interferometer, θ is the angle between the incoming light ray and the optical axis, δ is the phase change due to reflection (a constant term that can be ignored), m is an integer values representing the order of the interference, and λ is the wavelength.

In the static imaging mode, the mirror spacing (or thickness) and the refractive index of the Fabry-Perot interferometer remains constant while the angle of the incident light ray θ is varied by imaging points off-axis [3-13]. The fixed mirror spacing Fabry-Perot interferometer is also called as a Fabry-Perot etalon. An intensified CCD camera is usually used to record the interference patterns of the Fabry-Perot interferometer. A narrow linewidth monochromatic laser light pass through the Fabry-Perot etalon results in an image of the light source modulated in intensity by concentric light rings at locations associated with angles that meet the criterion of Eq.(8). The spectral of the received light scattered by the gas flow is broadened due to Rayleigh-Brillouin scattering, and the rings in the interference pattern are also broadened accordingly. The measurement of the rings's linewidth provides a means for deriving the gas temperature from the Rayleigh-Brillouin scattering spectra. The frequency shift of the light associated with the bulk flow velocity can be determined by the spatial shift in the rings' position on the interference pattern image.

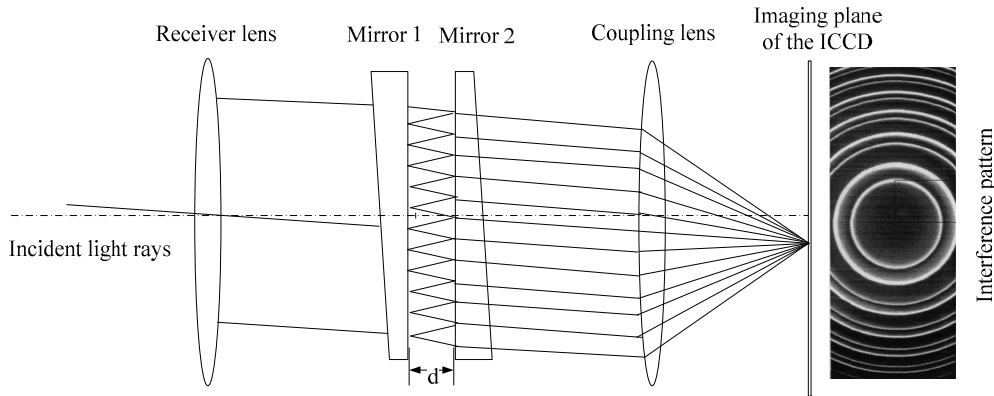


Fig. 2. Principle of Fabry-Perot interferometer.

The fringe intensity pattern is a function of the Rayleigh-Brillouin scattering spectrum convolved with the instrument function of the imaging Fabry-Perot interferometer. The instrument function of the imaging Fabry-Perot interferometer is given by

$$I_{IFPI}(\theta, \nu) = \int_{\theta_{\min}}^{\theta_{\max}} I_{FP}(\theta - \theta', \nu) I_{ICCD}(\theta) d\theta', \quad (9)$$

where $I_{IFPI}(\theta, \nu)$ is the instrument function of the imaging Fabry-Perot interferometer, $I_{FP}(\theta, \nu)$ is the instrument function of the Fabry-Perot etalon, $I_{ICCD}(\theta)$ is the instrument function of the intensified CCD camera, ν is the frequency, and θ is the angle between the incoming light ray and the optical axis.

The instrument function of the Fabry-Perot etalon is given by

$$I_{FP}(\theta, \nu) = \frac{I_0}{1 + F \sin^2[\psi(\theta, \nu)/2]}, \quad (10)$$

where $F=4R/(1-R)^2$ is the finesse of the Fabry-Perot etalon, R is the reflectivity of mirror, I_0 is the peak transmission of the Fabry-Perot etalon, $\psi(\theta, \nu)$ is the phase change. The phase change $\psi(\theta, \nu)$ is given by

$$\psi(\theta, \nu) = \frac{4\pi nd\nu}{c} \cos \theta. \quad (11)$$

Here n is the refractive index of the cavity medium of the Fabry-Perot etalon, d is the mirror spacing of the Fabry-Perot etalon, and c is the speed of light.

The instrument function (point scattering function, PSF) of the intensified CCD camera can be modeled as a Gaussian function:

$$I_{ICCD}(\theta) = \exp\left\{-\left(\frac{r^2}{\omega_{ICCD}^2}\right)\right\} = \exp\left\{-\left[\frac{(f \tan \theta)^2}{\omega_{ICCD}^2}\right]\right\}, \quad (12)$$

where $r = f \tan \theta$ is the radial position in the CCD image plane, f is the effective focal length of the coupling lens, ω_{ICCD} is the constant index of the point scattering function.

The expected digital number (DN) of the photoelectron counts of the ICCD camera can be calculated from [3-13]

$$\bar{N}_i(k, \theta) = \frac{\eta_{sys} n_{den} L_x \Delta t \left(\frac{\partial \sigma}{\partial \Omega}\right) \Omega_R P_0}{h\nu_L} \int_0^\infty I_{IFPI}(\theta, \nu) S(k, \nu - \nu') d\nu' + B_i \quad (13)$$

where $\bar{N}_i(k, \theta)$ is the DN of the measured Rayleigh-Brillouin scattering spectrum as a function of the interaction wave vector k and the angle θ , B_i is the DN of the background, η_{sys} is the efficiency of the imaging Fabry-Perot interferometer, L_x is the probe volume length, Δt is the gating time of the ICCD, $\partial \sigma / \partial \Omega$ is the differential scattering cross-section, Ω_R is the field of view (FOV) of the receiver lens, P_0 is the power of the incident laser beam, ν_L is the wavelength of the laser beam, h is the Plank constant, n_{den} is the molecular number density, $S(k, \nu)$ is the Rayleigh-Brillouin scattering spectrum.

4. ANALYSIS AND DISCUSSION

4.1 Resolving power of the imaging Fabry-Perot interferometer

The spectral resolving power quantifies the capability of the imaging Fabry-Perot interferometer to resolve two close wavelengths. It is defined as the ratio of the wavelength λ of the source to the minimum difference of wavelengths $\Delta\lambda_{min}$ that generates two circle series that can be discriminated. According to the Rayleigh's criterion, the spectral resolving power can be derived as

$$A = \left| \frac{\lambda}{\Delta\lambda_{min}} \right| \approx \frac{2\pi nd}{\lambda} \frac{\sqrt{R}}{1-R}, \quad (14)$$

where A is the spectral resolving power, m is the interference order, R is the reflectivity of the mirror. The minimum resolving wavelength is given by

$$\Delta\lambda_{min} = \frac{\lambda^2}{2\pi nd} \frac{1-R}{\sqrt{R}}. \quad (15)$$

Equation (15) can be written in the frequency space

$$\Delta \nu_{\min} = \frac{c}{2\pi nd} \frac{1-R}{\sqrt{R}}, \quad (16)$$

where $\Delta \nu_{\min}$ is minimum resolving frequency.

4.2 Free spectral range of the imaging Fabry-Perot interferometer

As is shown in Fig. (1), the Rayleigh-Brillouin scattering spectrum contains two Brillouin peaks, *i.e.*, the Stokes components that are red shifted and the anti-Stokes components that are blue shifted. Thus, there are three interference rings for each order of the imaging Fabry-Perot interferometer. In order to distinguish the Stokes component from the anti-Stokes component at a given order (the p^{th} order), one can get

$$FSR > 2\Delta \nu_{B_{\max}} + \Delta \nu_{-1/2}^p + \Delta \nu_{+1/2}^p = 2(\Delta \nu_{B_{\max}} + \Delta \nu_{1/2}^p), \quad (17)$$

where FSR is the free spectrum range of the imaging Fabry-perot interferometer, $\Delta \nu_{B_{\max}}$ is the maximum Brillouin frequency shift, $\Delta \nu_{-1/2}^p$ is the bandwidth (full width at half maximum, FWHM) of the anti-Stokes components, $\Delta \nu_{+1/2}^p$ is the FWHM of the Stokes component, and $\Delta \nu_{1/2}^p = (\Delta \nu_{-1/2}^p + \Delta \nu_{+1/2}^p) / 2$. Considering that $\Delta \nu_{B_{\max}} > \Delta \nu_{1/2}^p$, Eq. (17) can be written as

$$FSR > 3\Delta \nu_{B_{\max}}, \quad (18)$$

The spectral resolving power quantifies the capability of the imaging Fabry-Perot interferometer to resolve two close wavelengths or frequencies. From Eqs.(16) and (18), one can get

$$3\Delta \nu_{B_{\max}} < FSR < \frac{\pi\sqrt{R}}{(1-R)} \Delta \nu_{B_{\min}}. \quad (19)$$

The typical frequency shift of the Rayleigh-Brillouin scattering spectrum is in the range of 1-2GHz for gas flows. The refractive index of the solid Fabry-Perot etalon is 1.46 at the wavelength of 532nm. For solid Fabry-Perot etalon with mirror reflectivity of 0.8-0.9, the FSR of the imaging Fabry-Perot interferometer is in the range of 4.3GHz to 25.6GHz, and the thickness of the Fabry-Perot etalon is in the range of 4.0mm to 23.7mm.

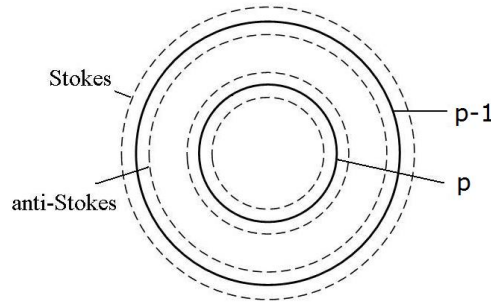


Fig. 3. Typical interference pattern of the imaging Fabry-Perot interferometer.

4.3 Field of view (FOV) of the receive optics

The accurate measurement of the focal length of the coupling lens in the Fabry-Perot interferometer is very difficult. Thus, it is bring a trouble to determent the angle between the incident ray and the optical axis accurately. The reconstruction algorithm for the Rayleigh-Brillouin scattering spectrum requires that several interferometer rings are necessary to eliminate the effect of the measurement error of the focal length. From Eq.(8), one can get

$$\theta^2 \approx 2 - \frac{mc}{ndv}, \quad (20)$$

where, θ is the angle of incidence in the gap of the Fabry-Perot etalon, m is the interference order, n is the refractive index, d is the thickness of the Fabry-Perot etalon, c is the speed of light, and ν is the frequency.

Considering that the interference order of the centering ring ($\theta=0$) is $m_0 = 2nd\nu/c$, the interference order of the p^{th} ring out from the center is $m - p$. The angle θ_p for the p^{th} ring is given by

$$\theta_p^2 = 2 - \frac{(m_0 - p)c}{nd\nu}. \quad (21)$$

Thus, the FOV of the receiver optics can be written as

$$FOV_p = 2n\theta_p = 2\sqrt{2n^2 - \frac{n(m_0 - p)c}{d\nu}}. \quad (22)$$

The FOV of the receiver optics is 2.47° or 2.85° for producing 3 or 4 interference rings, respectively.

5. CONCLUSION

Laser Rayleigh-Brillouin scattering provides an effective method for non-intrusive measurement of gas flow properties because that the laser Rayleigh-Brillouin scattering spectrum contains information about the density, temperature, and the bulk velocity of the gas flow. The measurement of these gas flow's properties relies on the exact knowledge of the spectral profile of the scattered light. The Fabry-Perot interferometer operated in the imaging mode can be used to resolve the Rayleigh-Brillouin scattering spectra, and the interference patterns of the Fabry-Perot interferometer is recorded by an intensified CCD camera for data process. The imaging Fabry-Perot interferometer has the advantage of high spectrum resolution providing information on the spectrum profile of the scattered laser light so that it is a powerful diagnostic tool for the study of gas flow properties.

REFERENCES

- [1] Lock, J. A., Seasholtz, R. G., and John, W. T., "Rayleigh-Brillouin scattering to determine one-dimensional temperature and number density profiles of a gas flow field," *Appl. Opt.*, 31(15), 2839-2848 (1992).
- [2] Anisimov, Y., Kosykh N., Mashek, I., and Smirnov, S., "Spectra of Rayleigh-Brillouin scattering in supersonic gas flows measured with a wide aperture spectrometer," *J. Phys.: Conf. Ser.*, 397(1), 012048 (2012).
- [3] Seasholtz, R. G., and Greer, L. C., "Rayleigh scattering diagnostic for measurement of temperature and velocity in harsh environments," *AIAA*, 1998-0206 (1998).
- [4] Seasholtz, R. G., and Panda, J., "Rayleigh scattering diagnostic for dynamic measurement of velocity and temperature," *AIAA*, 1999-0641 (1999).
- [5] Panda, J. and Seasholtz, R. G., "Velocity and temperature measurement in supersonic free jets using spectrally resolved rayleigh scattering," *AIAA*, 1999-0296 (1999).
- [6] Seasholtz, R. G. and Panda, J., "Rayleigh scattering diagnostic for simultaneous measurements of dynamic density and velocity," *AIAA*, 2000-0642 (2000).
- [7] Seasholtz, R. G., Panda, J., and Elam, K. A., "Rayleigh scattering diagnostic for measurement of velocity and density fluctuation spectra," *AIAA*, 2002-0827 (2002).
- [8] Mielke, A. F., Seasholtz, R. G., Elam, K. A., *et al.*, "Time-average molecular Rayleigh scattering technique for measurement of velocity, density, temperature, and turbulence intensity in high speed nozzle flows," *AIAA*, 2004-0706 (2004).
- [9] Mielke, A. F., Seasholtz, R. G., Elam, K. A., and Panda, J., "Time-average measurement of velocity, density, temperature, and turbulence velocity fluctuations using Rayleigh and Mie scattering," *Exp. Fluids*, 39(2), 441-454 (2005).

- [10] Mielke, A. F., Elam, K. A., and Sung, C. J., "Time-resolved Rayleigh scattering measurements in hot gas flows," AIAA, 2008-0262 (2008).
- [11] Mielke, A. F. and Elam, K. A., "Molecular Rayleigh scattering diagnostic for measurement of high frequency temperature fluctuations," Proc. SPIE 5880, (2005).
- [12] Mielke, A. F., Elam, K. A., and Clem, M. M., "Multiple-point mass flux measurement system using Rayleigh scattering," AIAA, 2009-528 (2009).
- [13] Mielke, A. F., Clem, M. M., Elam, K. A., and Hirt, S. M., "Progress on a Rayleigh scattering mass flux measurement technique," AIAA, 2010-856 (2010).
- [14] Miles, R. B., Lempert, W. R., and Forkey, J. N., "Laser Rayleigh scattering," Meas. Sci. & Technol., 12(5), R33-R51 (2001).
- [15] Ubachs, W. M. G, Duijn, van E. J., Vieitez, M. O., *et al.*, "A spontaneous Rayleigh-Brillouin scattering experiment for the characterization of atmospheric lidar backscatter," ESA Report, 1-203 (2010).
- [16] Zheng, Q., "On the Rayleigh-Brillouin scattering in air," PhD thesis, University of New Hampshire, 2004.
- [17] Tenti, G, Boley, C. D., and Desai, R. C., "On the Kinetic Model Description of Rayleigh-Brillouin Scattering from Molecular Gases," Canadian J. Phys., 52(4), 285-290(1974).
- [18] Vaughan, J. M., [The Fabry-Perot Interferometer: History, Theory, Practice, and Application], Taylor & Francis Group, New York, 1989.
- [19] Ma, J., Liu, B. Q., Hua, W. S., Luo, X. X., "Design of Fabry-Perot etalon with transmission linear to incident angle," J. Appl. Opt., 30(5), 869-873 (2009).